Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 10 · day per timetable	10	1 period (60 min)	Circles	Algebra II	Topic 9 — Conic Sections · Lesson 9-2	Grade 11 — 11B

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · Whiteboards · Graphing technology · Canva (final product) · Squared paper	Lesson presentation (15 slides) · "Choose Your Route" activity (3 pages) · Wayground kick-off

Competencies	Technology Embedded	Connection to student's real life
1- Writing the equation of a circle and reading its center and radius, including after completing the square. 2- Solving a line-circle system and verifying it on a graph.	1- Quizizz live diagnostic with instant gap analysis. 2- Graphing technology to check the intersection points of a line and a circle.	1- A plaza, a pool and a walkway: where does the path cross the water? 2- A radio tower: who is inside the covered area?

Values	Soft Skills	Hard Skills
1- Accuracy — every sign is written down before it is used. 2- Precision — exact answers first, decimals only at the end.	1- Collaboration in assigned group roles. 2- Explaining in words why the right-hand side is r^2.	1- Deriving $(x-h)^2+(y-k)^2=r^2$ from the distance formula. 2- Using a repeated root to show a tangent.

Lesson Integral Parts

Objectives / skills	Teacher Strategies & Student Activities	Assessment	Time
1- Derive the equation of a circle given its center and radius. 2- Complete the square to find the center and radius of a circle. 3- Determine the relative positions of a circle and a line in the plane. 4- Solve a system of equations with one linear equation and one equation of a circle algebraically and verify the solution by graphing. *Vocabulary: Center ; Circle ; Diameter ; radius.*	FIKR Phase 1 — Diagnose. 3-question Quizizz: completing the square, distance, solving a quadratic. Results read as a gap map and used to route students. FIKR Phase 2 — Targeted Instruction. The equation of a circle; completing the square; a line and a circle; solving the system. FIKR Phase 3 — Practice. Two identical "Give it a go" problems; then students choose a route and may switch. Compare routes and judge the claim. Pear Deck feedback. FIKR Phase 4 — Production. Hira Cultural District reflecting pool; AI as challenger only. FIKR Phase 5 — Proof of Learning. "Time to Check". No notes, no partner, no AI. FIKR Phase 6 — Smart Production. Design a circle and a line, find their intersections, swap and test; publish to the LMS portfolio.	Diagnostic quiz — gap map, not a grade. Whiteboard check (80% threshold) and "Give it a go" — self-marked. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. Final product graded on reasoning and clarity.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Two minutes of peer show-and-tell, then: "Which is harder — reading the signs in the equation, or deciding whether the line touches the circle?"
Reflection for Improvement:	
Homework:	Upload the final product to the LMS portfolio. Pear Deck practice set (12 questions, choose your route). Schoology homework quiz for Lesson 9-2. Watch the flipped video for the next lesson. Clinic group: bring your Time to Check paper.
Teacher name:	Mr Malek Thiab